

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO2: Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT3)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Report Analysis Questions

1. A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyze the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system. **(20 Marks)**

Commands :

1. nmap scanme.nmap.org
2. nmap -sV scanme.nmap.org
3. sudo nmap -O scanme.nmap.org
4. nmap -sS scanme.nmap.org
5. sudo nmap -A scanme.nmap.org

Observation :

- The host is alive.
- HTTP (80) is open.
- SSH (22) is open.
- SIP ports 5060 and 5061 are open.
- Operating system information identified.

Security Recommendations :

- Keep all software updated.
- Enable firewall protection.

- Use strong authentication mechanisms.
- Perform periodic vulnerability assessments.
- Conduct regular penetration testing.
- Use antivirus/anti-malware software.



Report : Question1_Nmap_Report.txt

```

kali@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ nmap --version
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.4 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.5 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select

kali@kali:~$ nmap scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:40 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0025s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
2022/tcp  closed down
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy

Nmap done: 1 IP address (1 host up) scanned in 88.31 seconds

kali@kali:~$ nmap -sV scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:43 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0084s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  tcpwrapped
53/tcp    closed domain
80/tcp    open  http        Apache httpd 2.4.7
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
8085/tcp  closed unknown

```

```

kali@kali: ~
Session Actions Edit View Help
kali@kali:~$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:49 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.011s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 985 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 9.8p1 Ubuntu 0ubuntu0.23.04.1 (Ubuntu 9.8p1-1ubuntu0.23.04.1)
80/tcp    open  http        Apache httpd 2.4.7 ((Ubuntu))
83/tcp    closed mt-ml-dev
139/tcp   closed netbios-ssn
143/tcp   closed imap
300/tcp   closed isakmp
2022/tcp  closed down
3128/tcp  closed squid-http
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy
OS fingerprint not ideal because: Didn't receive UDP response. Please try again with -sSU
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 38.05 seconds

kali@kali:~$ sudo nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:51 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0027s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http        Apache httpd 2.4.7 ((Ubuntu))
1 http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
Device type: bridge[101p] adapter[general purpose]
Running: (JUST GUESSED) Oracle Virtualbox (94%), Slirp (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:damny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (94%), AT&T BGM210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

```

```
Session Actions Edit View Help
Running (JUST GUESSING): Oracle Virtualbox (94%), Slirp (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (94%), AT&T BGM210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 1.77 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 94.05 seconds

(kali@kali)~$ sudo nmap -A scanme.nmap.org -oN Assignment1_Nmap_Report.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:54 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0020s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f83c:91ff:fe18:bb2f
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  tcpwrapped
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
53/tcp    closed domain
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   closed netbios-ssn
143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
2022/tcp  closed down
3389/tcp  closed ms-wbt-server
5660/tcp  open  sip?
5661/tcp  open  sip-tls?
8080/tcp  closed http-proxy
Device type: bridge|VoIP adapter|general purpose
Running (JUST GUESSING): Oracle Virtualbox (94%), Slirp (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (94%), AT&T BGM210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.40 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 93.15 seconds

(kali@kali)~$ ls
172.25.83.194 Assignment1_Nmap_Report.txt Desktop Documents Downloads ethical ethicalhacking index.html index.html.1 Music Pictures practical Public sample1.txt sample.txt Templates Videos
```

Risks

- Exposure of server information
- Credential theft
- Unauthorized system access

Conclusion

The Nmap Aggressive Scan successfully identified the target's IP address, open ports, running services, and probable operating system details. Such information can be valuable during the reconnaissance phase of an attack and may be used to conduct convincing social engineering attacks against IT support personnel. Implementing strong security controls, regular patch management, and employee awareness training can significantly reduce these risks.

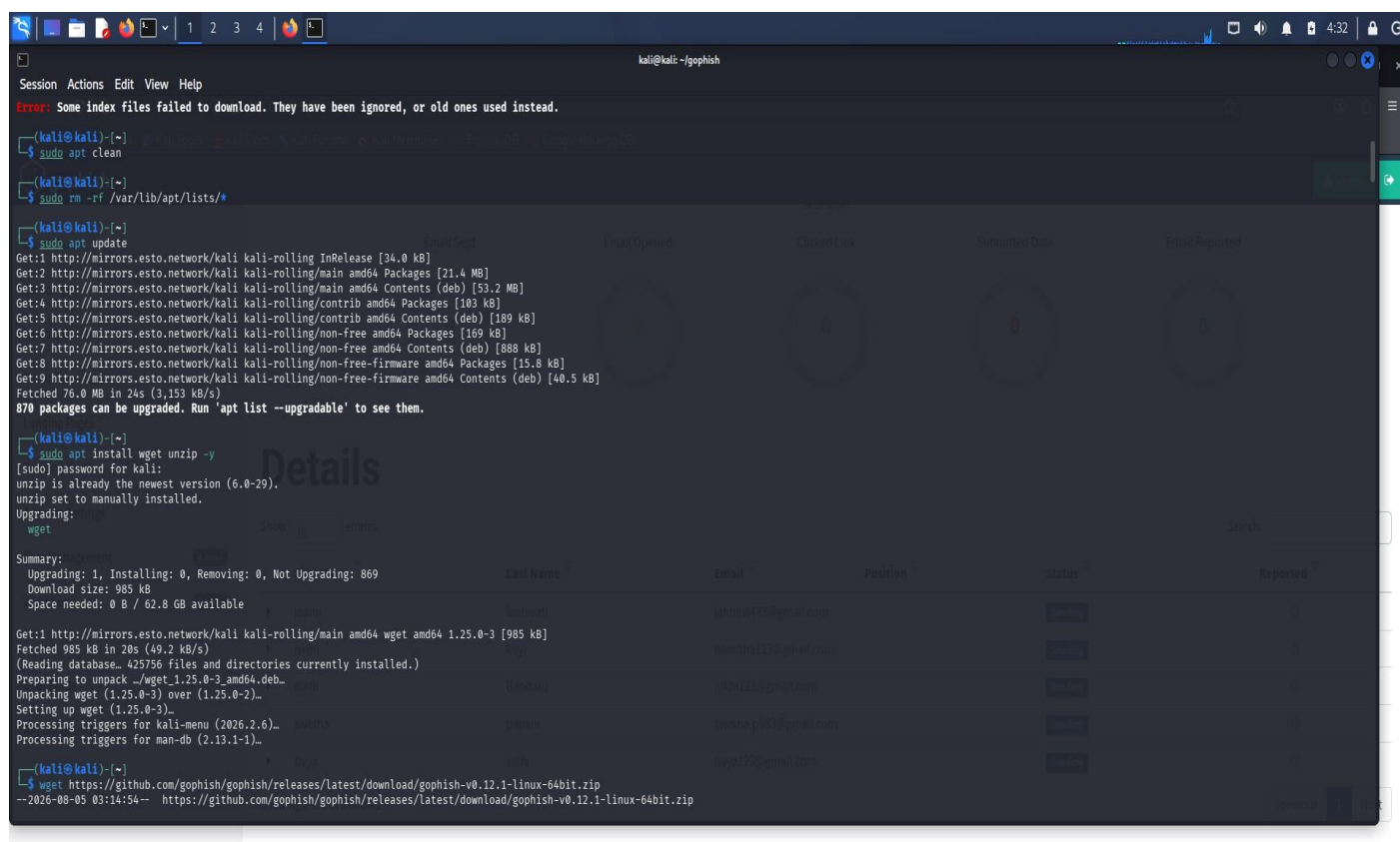
2. A phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. using the GoPhish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks.(20 Mark).

Sol : To conduct a phishing awareness simulation using **GoPhish** by creating an email template, sending phishing emails to dummy employee accounts, monitoring user interactions, and generating a report showing delivery rate, email open rate, and click rate. Finally, recommendations are provided to improve user awareness against phishing attacks.

Command 1 : sudo apt update

Purpose :

Updates the Kali Linux package repository to ensure the latest package information is available before installing software.



```
kali@kali: ~/gophish
Error: Some index files failed to download. They have been ignored, or old ones used instead.

(kali@kali)~$ sudo apt clean
(kali@kali)~$ sudo rm -rf /var/lib/apt/lists/*
(kali@kali)~$ sudo apt update
Get:1 http://mirrors.esto.network/kali kali-rolling InRelease [34.0 kB]
Get:2 http://mirrors.esto.network/kali kali-rolling/main amd64 Packages [21.4 MB]
Get:3 http://mirrors.esto.network/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]
Get:4 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Packages [103 kB]
Get:5 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Contents (deb) [189 kB]
Get:6 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Packages [169 kB]
Get:7 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Contents (deb) [888 kB]
Get:8 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Packages [15.8 kB]
Get:9 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Contents (deb) [40.5 kB]
Fetched 76.0 MB in 24s (3.153 kB/s)
870 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali)~$ sudo apt install wget unzip -y
[sudo] password for kali:
wget is already the newest version (6.0-29).
unzip is already the newest version (6.0-29).
unzip set to manually installed.
Upgrading:
  wget
Summary:
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 869
  Download size: 985 kB
  Space needed: 0 B / 62.8 GB available

Get:1 http://mirrors.esto.network/kali kali-rolling/main amd64 wget amd64 1.25.0-3 [985 kB]
Fetched 985 kB in 20s (49.2 kB/s)
(Reading database... 425756 files and directories currently installed.)
Preparing to unpack ./wget_1.25.0-3_amd64.deb...
Unpacking wget (1.25.0-3) over (1.25.0-2)...
Setting up wget (1.25.0-3)...
Processing triggers for kali-menu (2026.2.6)...
Processing triggers for man-db (2.13.1-1)...

(kali@kali)~$ wget https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip
--2026-08-05 03:14:54-- https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip
```

File Name	Email	Position	Status	Reported
gophish-v0.12.1-linux-64bit.zip	gophish@github.com	gophish	Downloaded	
gophish-v0.12.1-linux-64bit.zip	gophish@github.com	gophish	Downloaded	
gophish-v0.12.1-linux-64bit.zip	gophish@github.com	gophish	Downloaded	
gophish-v0.12.1-linux-64bit.zip	gophish@github.com	gophish	Downloaded	
gophish-v0.12.1-linux-64bit.zip	gophish@github.com	gophish	Downloaded	
gophish-v0.12.1-linux-64bit.zip	gophish@github.com	gophish	Downloaded	
gophish-v0.12.1-linux-64bit.zip	gophish@github.com	gophish	Downloaded	
gophish-v0.12.1-linux-64bit.zip	gophish@github.com	gophish	Downloaded	
gophish-v0.12.1-linux-64bit.zip	gophish@github.com	gophish	Downloaded	
gophish-v0.12.1-linux-64bit.zip	gophish@github.com	gophish	Downloaded	

Command 2 : wget https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip

Purpose :

Downloads the latest GoPhish Linux package from the official GitHub repository.

Command 3 : unzip gophish-v0.12.1-linux-64bit.zip -d gophish

Command 4 : cd gophish

Command 5 : chmod +x gophish

Command 5 : sudo ./gophish

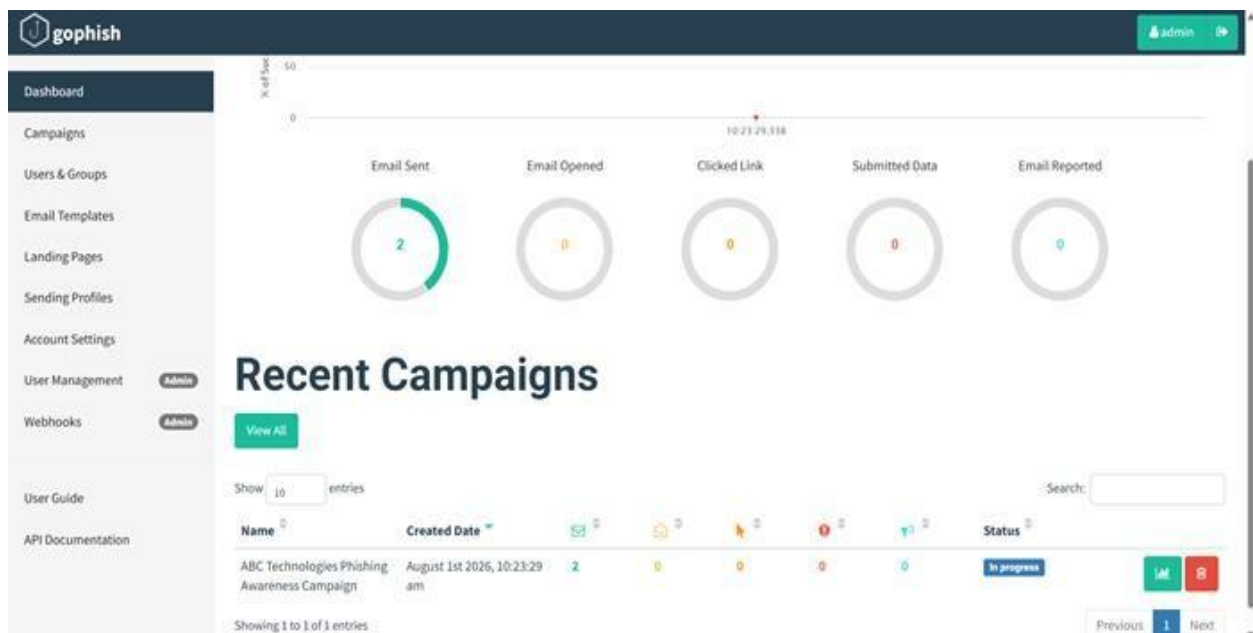
Purpose :

Extracts the downloaded GoPhish ZIP archive into the **gophish** directory. Changes the current working directory to the GoPhish installation folder. Grants execute permission to the GoPhish executable so that it can be started. Starts the GoPhish application, initializes the database, generates SSL certificates if required, and launches the phishing server and the administration web interface.



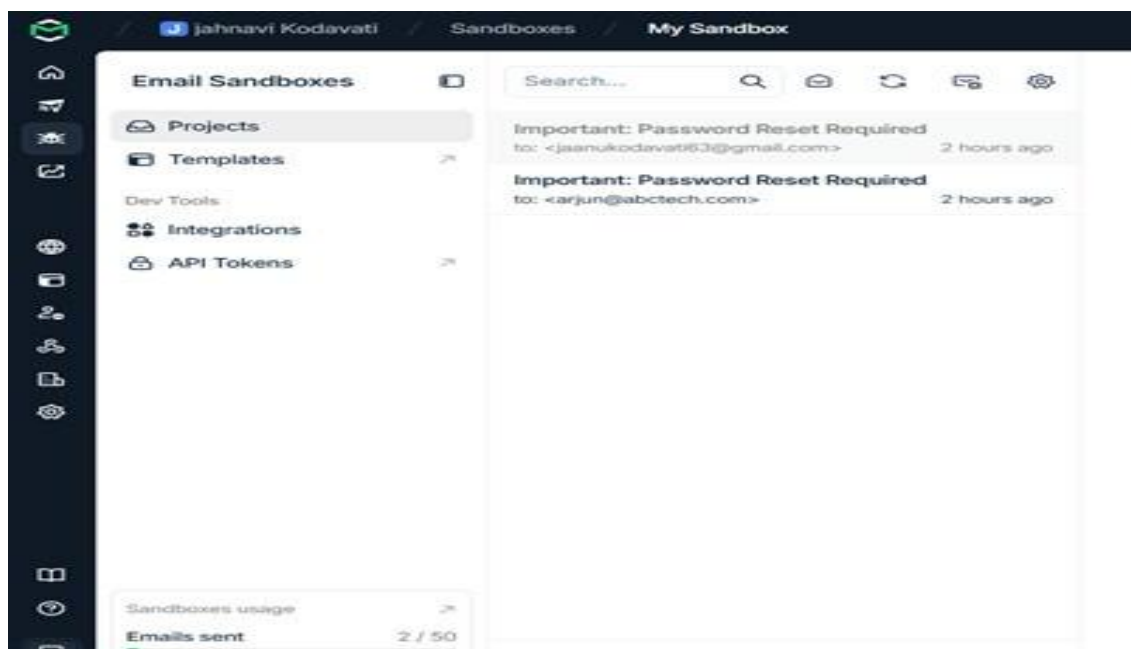
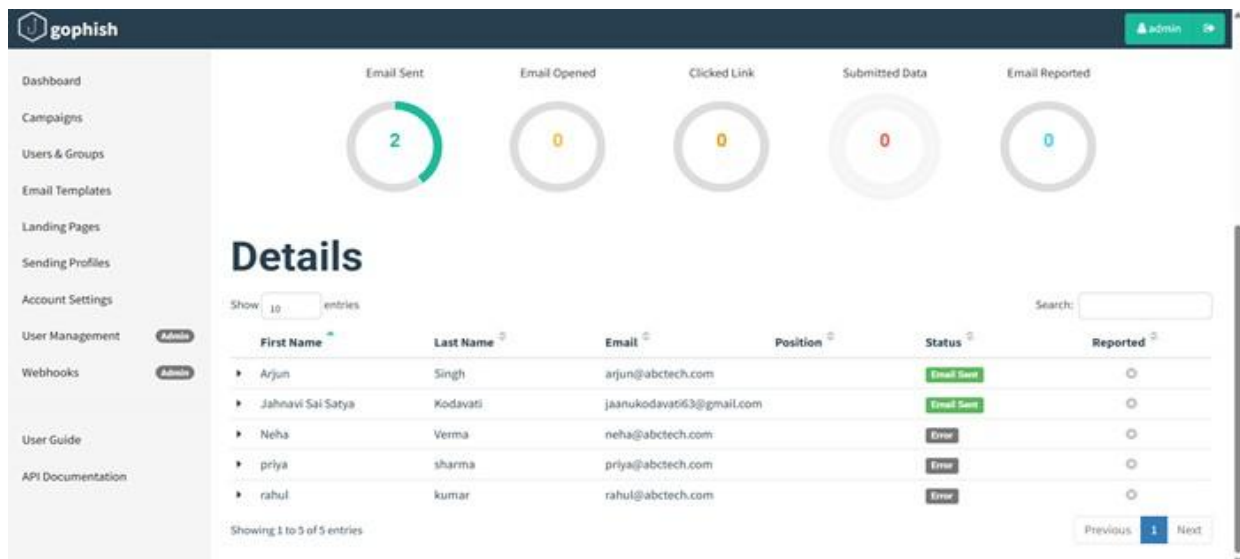
ABC Technologies
Phishing Awareness C

Report : <https://127.0.0.1:3333/>



The screenshot shows the 'Details' page in the GoPhish interface. The top navigation bar and left sidebar are identical to the dashboard. The main content area displays a table of users. The table has columns for First Name, Last Name, Email, Position, Status, and Reported. There are five users listed, all with a status of 'Pending'. The bottom of the page shows pagination: 'Showing 1 to 5 of 5 entries'.

First Name	Last Name	Email	Position	Status	Reported
jaanu	kodavali	jahmay483@gmail.com		Pending	
namo	koyl	namotha123@gmail.com		Pending	
nikhi	Bendaru	nikhi123@gmail.com		Pending	
sweetha	papari	sweetha.p993@gmail.com		Pending	
thiya	sathu	thiya123@gmail.com		Pending	



Email Security Controls :

- Enable Spam Filtering.
- Use Email Gateways.
- Configure SPF (Sender Policy Framework).
- Configure DKIM (DomainKeys Identified Mail).
- Configure DMARC (Domain-based Message Authentication, Reporting, and Conformance).
- Block dangerous attachments.
- Scan email attachments using antivirus software.
- Enable URL filtering

3. A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyze the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure..(10 Marks)

Command :

dig AXFR zonetransfer.me @nsztml.digi.ninja

Observation :

- DNS Zone Transfer succeeded.
- Internal DNS records were exposed.
- Mail server records were visible.
- Multiple hostnames and IP addresses were disclosed.
- DNS server configuration details were successfully enumerated.

Possible Attacks :

An attacker can use the exposed information to:

- Perform reconnaissance.
- Scan exposed servers.
- Launch phishing attacks.
- Target mail servers.
- Exploit vulnerable web servers.
- Attempt brute-force attacks on exposed services.
- Prepare for advanced cyberattacks.

To Secure DNS Infrastructure :

- Disable DNS Zone Transfer for unauthorized users.
- Allow AXFR only between trusted DNS servers.
- Configure Access Control Lists (ACLs).
- Enable DNSSEC to protect DNS integrity.
- Regularly update DNS software.
- Monitor DNS logs.
- Remove unnecessary DNS records.
- Perform regular DNS security audits.

Code of Conduct:

I J Sravani (192372141) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: J Sravani

RUBRICS FOR CALCULATION**Report Analysis**

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Report Analysis and Interpretation	25%	Accurately interprets the security report, identifies all key findings, and explains their significance with excellent technical understanding.	Correctly interprets most findings with minor omissions or inaccuracies.	Identifies only some findings and provides limited interpretation.	Unable to interpret the report or identify key findings.
Identification of Security Risks and Vulnerabilities	25%	Correctly identifies all security risks, vulnerabilities, and possible attack vectors with clear technical justification.	Identifies most risks and vulnerabilities with minor errors.	Identifies only a few risks with limited technical explanation.	Fails to identify major risks or vulnerabilities.
Application of Cybersecurity Concepts and Countermeasures	30%	Applies appropriate ethical hacking concepts and recommends comprehensive, practical, and technically sound countermeasures.	Recommends suitable countermeasures with minor technical gaps.	Provides limited or partially appropriate recommendations.	Recommendations are inappropriate, incomplete, or missing.
Technical Communication and Documentation	20%	Response is well-organized, technically accurate, clearly presented, and uses appropriate cybersecurity terminology throughout.	Response is organized and understandable with minor documentation or presentation issues.	Response lacks clarity, organization, or contains several technical inaccuracies.	Response is poorly organized, incomplete, and difficult to understand.